Scientific Sessions

Predicting Heart Disease Using Machine Learning Classification Models

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American Heart Association.

#AHA25

Business Problem, Background, Objectives

Business Problem: Heart disease is the leading cause of death globally, claiming 17.9 million lives annually. Early detection is critical but often delayed, resulting in preventable deaths and escalating healthcare costs. Traditional risk assessment relies heavily on physician judgment, which can vary in accuracy.

Background: The Cleveland Heart Disease dataset contains clinical measurements from 297 patients, including demographics, vital signs, and diagnostic test results. This data provides an opportunity to build predictive models that can assist healthcare providers in identifying at-risk patients earlier.

Project Objectives:

1. Develop an accurate classification model to predict heart disease presence
2. Compare multiple machine learning techniques to identify the best-performing approach
3. Provide actionable insights on key risk factors
4. Deliver a reliable clinical decision-support tool that can improve patient outcomes and optimize resource allocation

Dataset Overview

Dataset Characteristics: (More in appendix)

- **Total Records:** 297 patients
- **Training Set:** 207 patients (70%)
- **Testing Set:** 90 patients (30%)
- **Target Variable:** Condition (0 = No Disease, 1 = Disease Present)
- **Class Distribution:** 46% without disease, 54% with disease (relatively balanced)

Key Predictor Variables (13 Features):

- **Demographic:** Age, Sex
- **Clinical Measurements:** Resting Blood Pressure (trestbps), Cholesterol (chol), Maximum Heart Rate (thalach)
- **Diagnostic Tests:** Chest Pain Type (cp), Fasting Blood Sugar (fbs), Resting ECG Results (restecg), Exercise-Induced Angina (exang)
- **Advanced Metrics:** ST Depression (oldpeak), Slope of ST Segment, Number of Major Vessels (ca), Thalassemia Type (thal)

Data Exploration (more in appendix) + Processing

Data Quality Assessment:

- Complete dataset with no missing values across all 297 patient records
- All 13 clinical features retained for modeling after validation
- Applied StandardScaler normalization to continuous variables (age, blood pressure, cholesterol, heart rate, ST depression)

Target Variable Distribution:

- **Class 0 (No Disease):** 160 patients (54%)
- **Class 1 (Disease Present):** 137 patients (46%)
- Dataset is well-balanced, minimizing class imbalance issues that could bias model predictions

Data Preprocessing Steps:

- Train-test split: 70-30 ratio (207 training, 90 testing) with random_state=42 for reproducibility
- Feature scaling applied to ensure distance-based algorithms (K-NN) perform optimally
- All 13 features retained based on clinical relevance and correlation analysis

Machine Learning Techniques Applied

Model 1: Logistic Regression

- **What it does:** Calculates the probability a patient has heart disease based on their clinical measurements
- **Key strength:** Easy to interpret—shows exactly how much each factor (age, cholesterol, etc.) increases or decreases disease risk
- **Trade-off:** Works best when relationships are straightforward; may miss complex patterns

Model 2: K-Nearest Neighbors (K-NN)

- **What it does:** Classifies each patient by looking at the 13 most similar patients in the training data
- **Key strength:** Naturally captures complex patterns without making assumptions about the data
- **Trade-off:** Requires careful tuning to find the right number of neighbors (K=13 in our case)

Model 3: Random Forest

- **What it does:** Creates 200 decision trees that each "vote" on whether a patient has disease; majority wins
- **Key strength:** Highly accurate and handles complex relationships well; resistant to overfitting
- **Trade-off:** Functions as a "black box"—harder to explain exactly why a prediction was made

Model 4: XGBoost (Gradient Boosting)

- **What it does:** Builds decision trees sequentially, with each new tree learning from the previous tree's mistakes
- **Key strength:** State-of-the-art performance in many competitions; includes built-in safeguards against overfitting
- **Trade-off:** Requires more fine-tuning; can be sensitive to parameter choices

Comparative Model Results: Accuracy Analysis

Performance Summary Table:

Model	Accuracy	Precision (Class 0)	Precision (Class 1)	Recall (Class 0)	Recall (Class 1)	F1-Score
K-NN	**84.4%**	85%	83%	85%	83%	0.84
Random Forest	83.3%	84%	83%	85%	81%	0.83
XGBoost	80.0%	81%	79%	81%	79%	0.80
Logistic Regression	78.9%	82%	76%	77%	81%	0.79

Discriminatory Power: AUC-ROC Comparison

Rank	Model	AUC Score	Interpretation
1	K-NN	0.844	Excellent discrimination
2	Random Forest	0.832	Excellent discrimination
3	Logistic Regression	0.790	Acceptable discrimination
4	XGBoost	0.799	Acceptable discrimination

Logistic Regression: Interpretable Risk Factors

Feature	Odds Ratio	Interpretation
ca (# vessels)	4.05	Each additional vessel colored increases disease odds by 305%
sex	2.62	Males have 162% higher odds of disease
thal	2.26	Thalassemia abnormality increases odds by 126%
trestbps	1.86	Higher blood pressure increases odds by 86%
oldpeak	1.70	Greater ST depression increases odds by 70%
thalach	0.48	Higher max heart rate reduces odds by 52%

Understanding Model Predictions: Where Do Models Succeed and Fail?

Comparison Across Models:

Model	True Negatives	False Positives	False Negatives	True Positives
K-NN	41	7	7	35
Random Forest	41	7	8	34
Logistic Regression	37	11	8	34
XGBoost	39	9	9	33

Final Recommendations & Business Impact

Winner: K-Nearest Neighbors (K=13) Based on comprehensive evaluation across multiple metrics, K-NN emerges as the superior model for heart disease prediction:

- **Highest accuracy:** 84.4%
- **Best AUC score:** 0.844
- **Balanced precision-recall:** 84% macro average
- **Fewest false negatives:** Only 7 missed diagnoses

Business Recommendations:

1. **Implement K-NN model** as primary screening tool in clinical decision support systems
2. **Deploy Random Forest as backup** (83.3% accuracy) for validation and ensemble approaches
3. **Use Logistic Regression for interpretation:** Despite lower accuracy (78.9%), its odds ratios provide valuable clinical insights into risk factors
4. **Focus interventions on high-risk indicators:** Number of vessels (ca), sex, and thalassemia type

THANK YOU

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Appendix A1: Dataset Sample

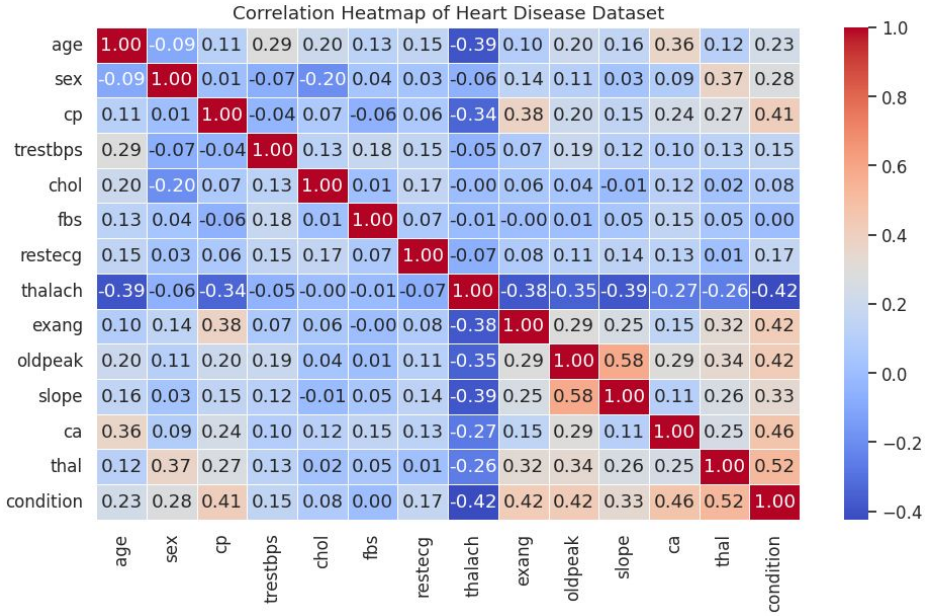
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	condition
0	69	1	0	160	234	1	2	131	0	0.1	1	1	0	0
1	69	0	0	140	239	0	0	151	0	1.8	0	2	0	0
2	66	0	0	150	226	0	0	114	0	2.6	2	0	0	0
3	65	1	0	138	282	1	2	174	0	1.4	1	1	0	1
4	64	1	0	110	211	0	2	144	1	1.8	1	0	0	0

Appendix B1: EDA Results, Predictors of Heart Disease

Key Findings from Correlation Analysis:

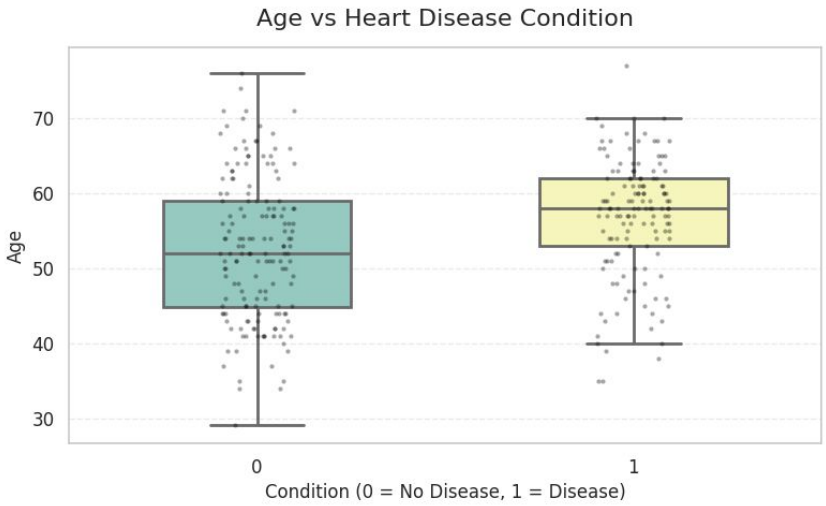
Strongest Predictors of Heart Disease (from correlation heatmap):

- **thal (Thalassemia):** +0.52 correlation - strongest single predictor
- **ca (# of major vessels):** +0.46 correlation - second strongest indicator
- **thalach (Max heart rate):** -0.42 correlation - negative relationship (protective effect)
- **cp (Chest pain type):** +0.41 correlation - significant diagnostic indicator
- **exang (Exercise-induced angina):** +0.42 correlation - important symptom marker
- **oldpeak (ST depression):** +0.42 correlation - critical ECG measurement
- **slope (ST segment slope):** +0.33 correlation - moderate predictor



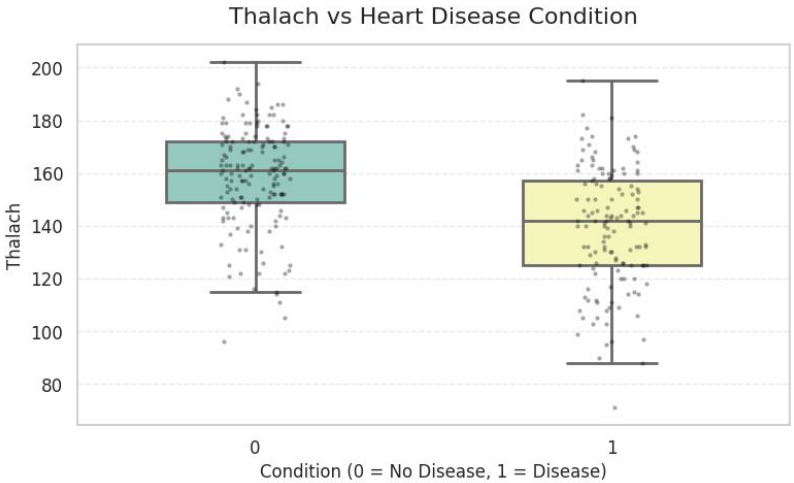
Appendix B2: EDA Results, Age Distribution

- **Patients without disease:** Median age ~52 years, range 29-77 years
- **Patients with disease:** Median age ~58 years, range 35-77 years, with outliers
- Heart disease patients tend to be slightly older, though substantial overlap exists across age groups



Appendix B3: EDA Results, Maximum Heart Rate

- **No disease patients:** Median ~160 bpm (higher maximum heart rates during exercise)
- **Disease patients:** Median ~141 bpm (significantly lower maximum heart rates)
- **Clinical insight:** Reduced exercise capacity (lower max heart rate) is a strong indicator of cardiovascular compromise

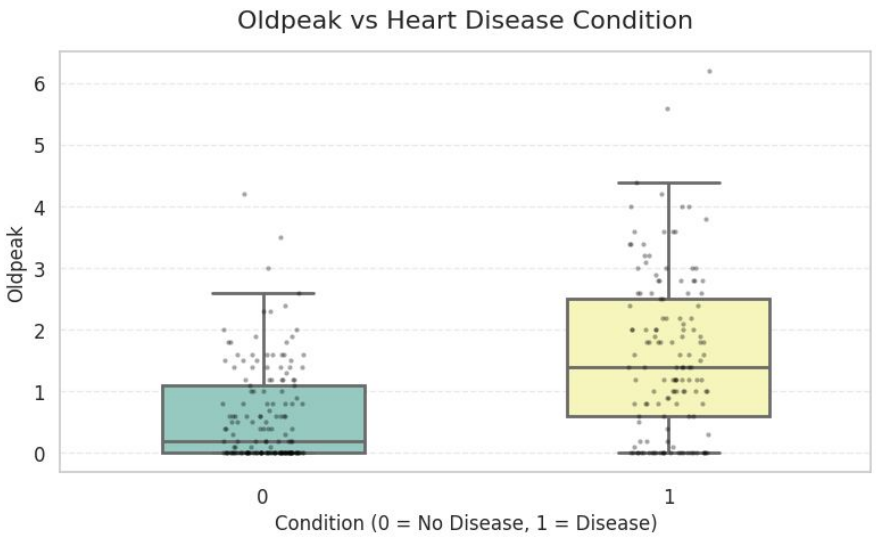


Appendix B4: EDA Results, ST Depression - oldpeak

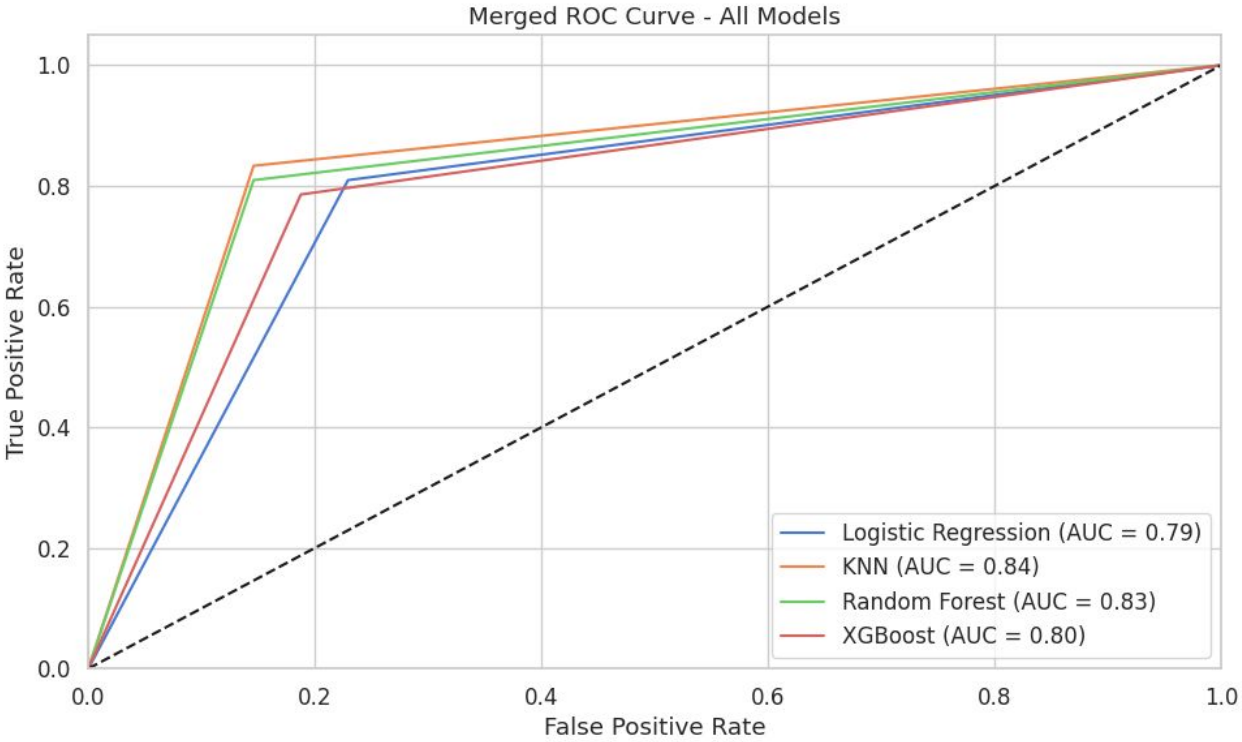
No disease patients: Median ~0.3, mostly concentrated near zero

Disease patients: Median ~1.5, with wider distribution up to 6.0

Clinical insight: Greater ST segment depression during exercise indicates insufficient oxygen supply to heart muscle, a hallmark of coronary artery disease

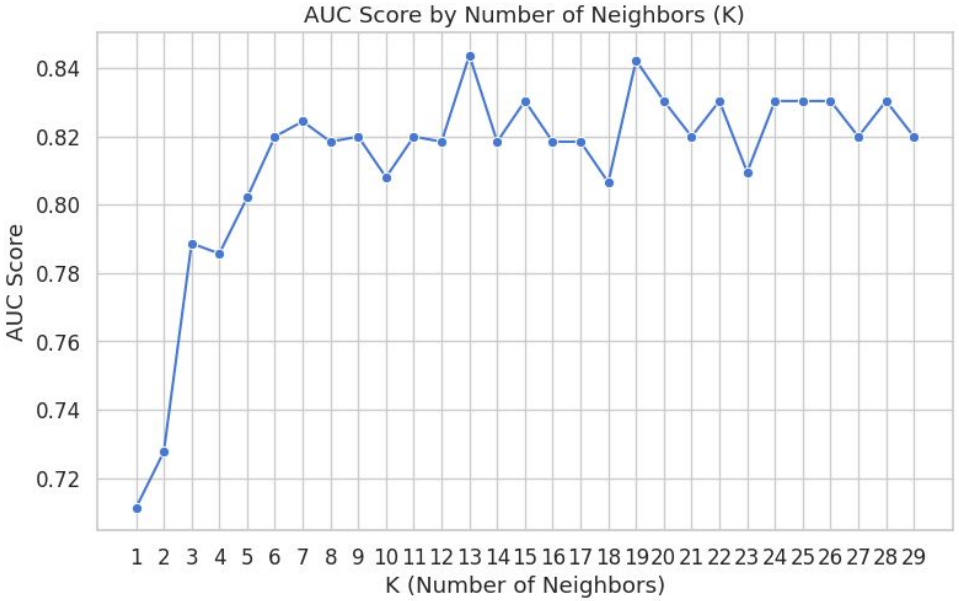


Appendix C: ROC Curves

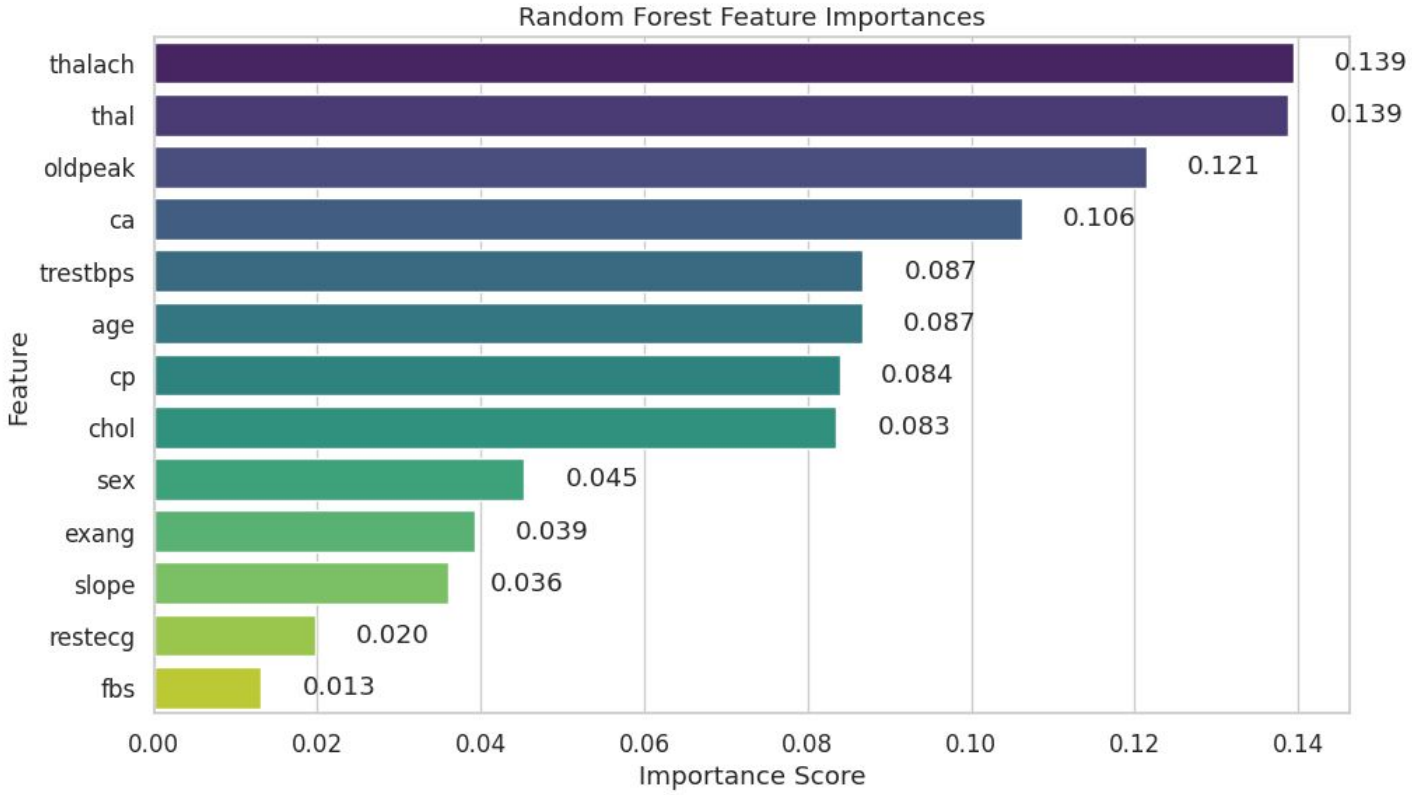


Appendix D: K-Selection Process

- Systematically tested K from 1-29 neighbors using AUC as the selection criterion
- **K=13 yielded the highest AUC score (0.844)** on test data
- This optimal K balances model complexity, avoiding both overfitting (low K) and underfitting (high K)



Appendix E: Feature Importance (Random Fore



Appendix E: EDA Results, ST Depression - oldpeak

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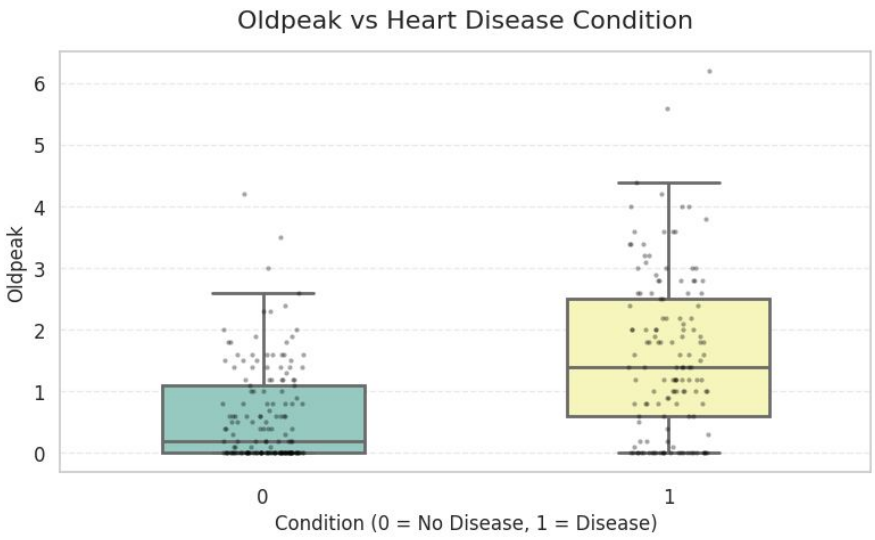
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